Design Thinking Case Study Report

First Three Stages: Empathize, Define, Ideate

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Topic: Unsafe Closing Doors in Mumbai Local Trains

1. Introduction

Design Thinking is a user-centered approach to problem solving that focuses on understanding the needs, behaviors, and experiences of users before developing solutions. It is widely used in areas such as product design, transportation systems, user experience, and urban planning. Unlike traditional methods that focus directly on technical fixes, Design Thinking first emphasizes understanding real-world problems faced by people.

The Design Thinking process generally includes five stages: Empathize, Define, Ideate, Prototype, and Test. However, in this case study, we will focus only on the first three stages: Empathize, Define, and Ideate. These stages are important because they help in deeply understanding the problem and generating effective solutions before implementation.

This case study focuses on the problem of unsafe door conditions in Mumbai local trains. Mumbai local trains are one of the busiest railway networks in the world and serve millions of passengers daily. Due to extreme overcrowding and lack of automated door systems, passengers often travel while standing near open doors, which leads to serious safety risks.

Passengers frequently fall from trains, get injured while boarding or exiting, or get pushed due to crowd pressure near doors. Therefore, this case study aims to apply Design Thinking to understand this issue and develop a user-centered solution that improves safety while maintaining efficiency

2. Understanding the Problem Context

In Mumbai, local trains are the backbone of daily transportation. Millions of people rely on them for commuting to work, college, and other activities. However, due to limited train capacity and high population density, trains are extremely crowded, especially during peak hours.

One of the major safety issues in Mumbai local trains is that doors remain open while the train is in motion. Unlike metro systems, most local trains do not have automatic doors. As a result, passengers stand near open doors to get in and out quickly, which increases the risk of accidents.

Many passengers prefer standing near doors because they need to exit quickly at their station. However, due to overcrowding, they are often pushed or lose balance. This leads to cases where passengers fall from moving trains. Additionally, while boarding, people sometimes try to enter before others exit, causing chaos and increasing injury risks.

Railway authorities face challenges such as managing crowd flow, ensuring safety, and maintaining train schedules. Implementing strict rules is difficult because of the high volume of passengers and lack of enforcement.

This problem is not only technical but also behavioral and emotional. Passengers feel rushed, stressed, and sometimes helpless due to overcrowding. Therefore, any solution must consider both safety and user behavior.

3. Stage 1 – Empathize

The Empathize stage focuses on understanding the users, their experiences, and their challenges. Instead of making assumptions, real-world observations and interactions are used to gather insights.

To understand this problem, different methods can be used such as:

- Observing passengers during peak hours
- Interviewing daily commuters
- Studying accident reports
- Talking to railway staff

During observation, it is seen that passengers often stand at the edge of train doors due to lack of space inside. Many commuters are used to this behavior and consider it normal, even though it is unsafe.

Interviews reveal that passengers prioritize reaching their destination on time over safety. Many users say they have no choice but to stand near doors because of overcrowding. Some passengers also mention that they feel unsafe but cannot change their travel habits.

Railway staff highlight that controlling crowd behavior is very difficult. Even when warnings are given, passengers often ignore them due to urgency.

Emotionally, passengers experience stress, fear, and frustration. Some feel anxious while traveling near doors, while others feel forced into unsafe situations. At the same time, they want faster boarding and exiting.

From the empathy stage, the following key insights were obtained:

- Passengers stand near doors due to overcrowding
- Many passengers are aware of risks but still take them
- Lack of automatic doors increases danger
- Behavioral habits play a major role in the problem
- Passengers want both safety and convenience
- Railway authorities struggle to enforce safety rules

These insights help in clearly defining the problem in the next stage.

4. Stage 2 – Define

The Define stage involves analyzing the information collected and identifying the core problem.

Based on the empathy research, the main issue is that passengers traveling in Mumbai local trains are exposed to serious safety risks due to open doors and overcrowding, which leads to accidents and injuries.

The problem statement can be written as:

Passengers in Mumbai local trains need a safe and efficient system for managing train doors because overcrowding and open doors lead to unsafe travel conditions, accidents, and lack of control over passenger movement.

In this stage, user needs are also identified. Passengers need:

- Safe boarding and exiting
- Proper crowd management
- Clear alerts and instructions

- A system that does not delay travel

Railway authorities need:

- Better control over passenger movement
- Reduced accidents
- Efficient implementation without slowing trains

To explore solutions, “How Might We” questions are created:

- How might we ensure doors close safely before train movement?
- How might we reduce overcrowding near doors?
- How might we influence passenger behavior for safer travel?
- How might we maintain speed while improving safety?
- How might we design a system suitable for high crowd conditions?

These questions guide the ideation stage.

5. Stage 3 – Ideate

The Ideate stage focuses on generating multiple creative solutions without limiting ideas.

Several ideas were generated during brainstorming:

One idea is installing fully automatic sliding doors like metro trains. This ensures doors remain closed during movement. However, implementation cost and infrastructure changes are challenges.

Another idea is using sensor-based systems that detect obstacles and prevent doors from closing if someone is in the way. This improves safety but still requires behavioral adaptation.

A third idea is using audio and visual alerts such as loud announcements and flashing lights before door closing. This prepares passengers in advance.

Another idea is AI-based crowd monitoring using cameras to detect overcrowding near doors and send alerts to train operators.

A more practical solution is a Smart Automated Door Safety System. This system includes semi-automatic doors that close before train movement, combined with sensors and warning systems.

The system would work as follows:

- Doors remain open at stations
- Before train departure, alerts are given
- Doors close automatically with sensors for safety
- Visual indicators show door status
- Emergency override is available

This solution balances safety, cost, and practicality.

6. Proposed Solution Description

The proposed solution is a Smart Automated Train Door Safety System designed for Mumbai local trains.

The system includes automatic or semi-automatic doors integrated with sensors and alert systems. Before the train starts, an announcement and warning signal is given. After a few seconds, doors close automatically.

Sensors ensure that doors do not close if a passenger is blocking them. LED indicators show whether doors are open or closed. Emergency buttons are available for safety situations.

This system reduces accidents, improves crowd control, and enhances passenger confidence while traveling.

7. Conclusion

This case study applied the first three stages of Design Thinking—Empathize, Define, and Ideate—to address the issue of unsafe doors in Mumbai local trains.

The empathy stage helped in understanding passenger behavior and challenges. The define stage clarified the core problem and user needs. The ideate stage generated multiple solutions and identified the most practical one.

The final solution focuses on improving safety while maintaining efficiency, making train travel safer for millions of passengers.

This case study shows how Design Thinking can be used to solve real-world urban problems by focusing on users and their experiences.